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# Model Evaluation

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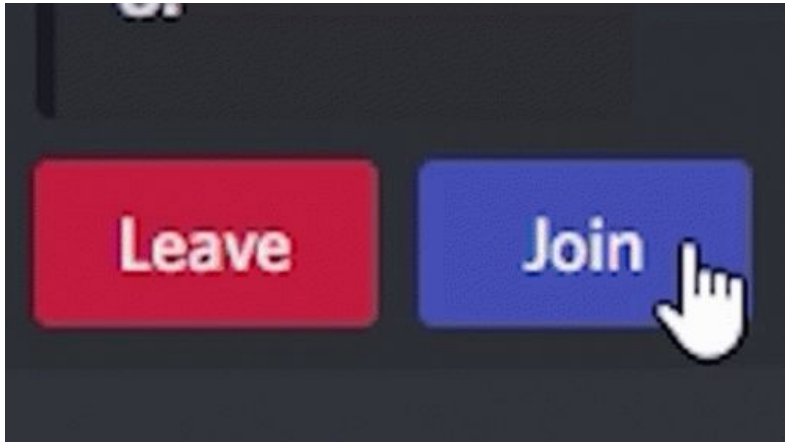

# Agenda

- Why Evaluate?
- Train vs. Test & Cross-Validation
- The Evaluation Framework
- Regression Metrics (MAE, MSE, RMSE, $R^2$)
- Visual Evaluation (Residuals)
- Classification Metrics (Accuracy, Precision, Recall)
- F1-Score & Log Loss
- Summary & Best Practices

# Why Evaluate?

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## Business Impact

A "90% accurate" model can still lose money. In fraud detection, missing a single high-value fraud case (False Negative) is far costlier than flagging a legitimate transaction.

## Technical Reality

Evaluation metrics help distinguish true signal from noise. They provide the quantitative evidence needed to trust a model's predictions on unseen data.

# Data Splitting Strategies

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### Train vs. Test

Separating data into training and testing sets is crucial to simulate how the model performs on unseen real-world data.

### Cross-Validation

K-Fold Cross-Validation ensures statistical robustness by training and testing on multiple subsets of the data.

### Holdout Sets

A final holdout set acts as a vault, preventing data leakage during hyperparameter tuning.

# The Evaluation Framework

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## Regression

**Goal:** Minimize distance between predicted and actual values.

**Predicts:** Continuous values.

*Examples: House Prices, Temperature, Stock Value.*

## Classification

**Goal:** Correctly assign class labels and probabilities.

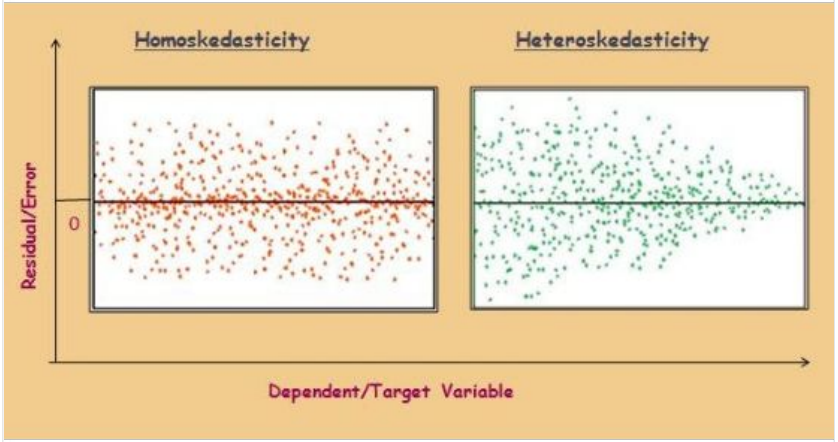
**Predicts:** Discrete labels.

*Examples: Spam/Ham, Disease/Healthy, Churn/Retain.*

# Introduction to Regression Error

## Understanding Residuals

- The core concept of regression error is the **Residual**.
- Defined as the difference between the actual $y$ value and the predicted $\hat{y}$ value .
- **Goal:** Find the line that minimizes the aggregate of these vertical distances.



# Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^{n} | y_i - \hat{y}_i |$$

The average magnitude of errors in a set of predictions, without considering their direction.

## Pros & Cons

- **Pros:** Robust to outliers; easy to interpret as it is in the same units as the output variable.
- **Cons:** Doesn't penalize large errors heavily. A big mistake is treated linearly compared to a small one.

# Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

The average of the squares of the errors.

## Key Characteristic

- **Penalty:** Squaring the error punishes large mistakes much more severely than small ones.
- **Cons:** The metric is not in the original units of the data (units are squared), making direct interpretation difficult.

# Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

The standard deviation of the prediction errors.

## Why use it?

- Brings the error metric back to the original units (like MAE).
- Retains the high penalty for large errors from MSE.
- **Comparison:** RMSE is always ≥ MAE. The difference indicates variance in the individual errors.

# Coefficient of Determination ($R^2$)

## Goodness of Fit

**Definition:** The proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

## Interpretation

- An $R^2$ of 0.80 means the model explains 80% of the variability.
- Range is typically 0 to 1.
- Useful for explaining model power to non-technical stakeholders.

# Adjusted $R^2$

## The Problem

Standard $R^2$ artificially increases as you add more features to the model, even if those features are useless noise. This encourages overfitting.

## The Solution

Adjusted $R^2$ introduces a penalty term for the number of predictors.

- **Use Case:** Critical when comparing models with different numbers of features.
- If you add a feature and Adjusted $R^2$ drops, the feature is likely not useful.

# Mean Absolute Percentage Error

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Error represented as a percentage.

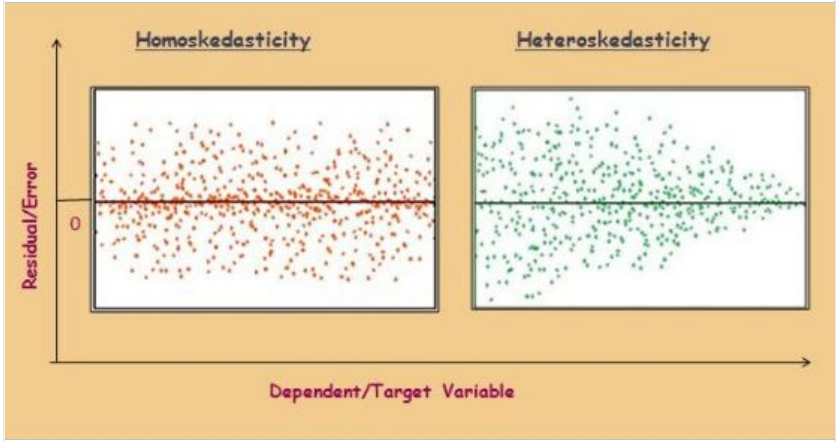
## Interpretation

- **Pros:** Scale-independent, making it excellent for comparing performance across different datasets.
- **Cons:** Undefined if actual value $y = 0$. It also punishes negative errors differently than positive ones.

# Visual Evaluation: Residual Plots

## Reading the Patterns

- **Ideal:** A random cloud of points around 0. This indicates homoscedasticity (equal variance).
- **Bad:** Patterns like a U-shape or a funnel shape.
- **Diagnosis:** Funnel shapes indicate heteroscedasticity, meaning the model's error grows with the value size.



# Choosing the Right Metric

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### Use MAE

When outliers are not critical business failures. It provides a robust, easy-to-understand average error.

### Use RMSE

When large errors are disastrous (e.g., predicting safety margins). The squaring term penalizes big deviations.

### Use $R^2$

To understand "goodness of fit" relative to data variance. Useful for explaining model power to stakeholders.

# Introduction to Classification

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## The Accuracy Trap

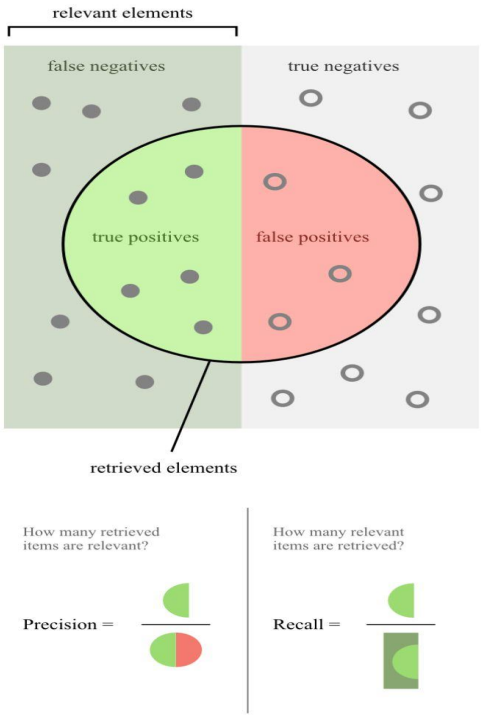
Why accuracy is not enough: The problem of **Imbalanced Datasets**.

- **Example:** In a dataset with 99% healthy and 1% sick patients, a model that predicts "Healthy" for everyone is 99% accurate but useless.
- We need to measure Type I (False Positive) and Type II (False Negative) errors specifically.

# The Confusion Matrix

## Foundational Table

- **True Positive (TP):** Correctly predicted positive.
- **True Negative (TN):** Correctly predicted negative.
- **False Positive (FP):** Incorrectly predicted positive (Type I Error).
- **False Negative (FN):** Incorrectly predicted negative (Type II Error).



# Accuracy

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

## Usage & Limitations

- **When to use:** Only when classes are balanced (e.g., 50/50 split).
- **The Trap:** It masks performance on minority classes. A high accuracy score can hide a model's complete inability to detect the event of interest.

# Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

**Focus:** Minimizing False Positives.

## Interpretation

**Question Answered:** "Of all the instances the model predicted as positive, how many were actually positive?"

**Example:** Spam detection. You don't want to label an important email as spam (False Positive), so high precision is key.

# Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN}$$

**Focus:** Minimizing False Negatives.

## Interpretation

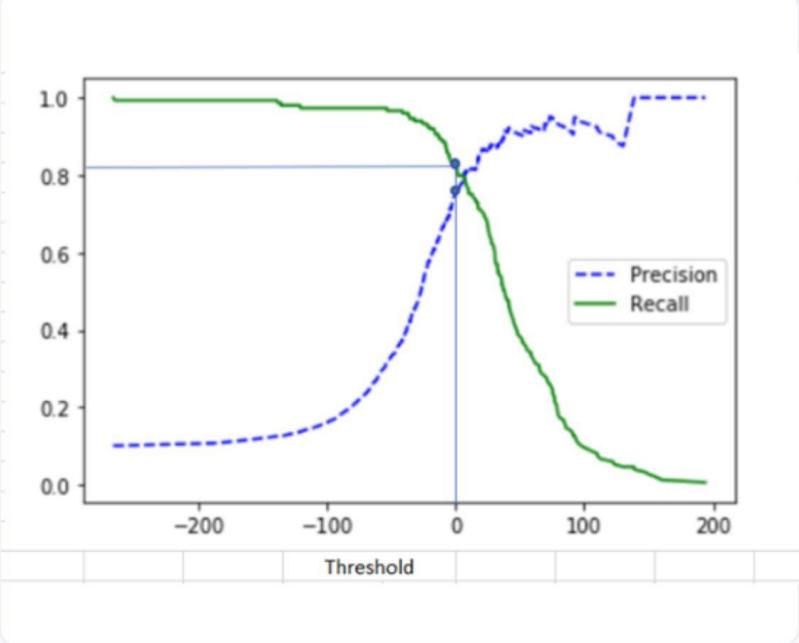
**Question Answered:** "Of all the actual positive instances, how many did the model capture?"

**Example:** Cancer diagnosis. You cannot afford to miss a sick patient (False Negative), so high recall is critical.

# The Precision-Recall Trade-off

## Inverse Relationship

- There is often a tension between Precision and Recall.
- **High Threshold:** Model is conservative. Few predictions, but likely correct. → High Precision, Low Recall.
- **Low Threshold:** Model is liberal. Captures everyone, including noise. → High Recall, Low Precision.



# F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Harmonic Mean of Precision and Recall.

## Why Harmonic Mean?

It punishes extreme values. If either Precision or Recall is 0, the F1-Score is 0.

**Use Case:** The best single metric for imbalanced datasets when you care about both False Positives and False Negatives.

# Advanced Metrics

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## Log Loss

Heavily penalizes confident predictions that are wrong. Predicting 0.99 probability for an event that does not happen incurs a massive penalty.

It forces the model to be honest about its uncertainty.

## Cohen's Kappa

A statistic that measures inter-rater agreement for categorical items.

**Meaning:** It tells you how much better your model is performing compared to a random guesser. A Kappa of 0 means the model is no better than chance.

# Summary & Best Practices

| Type | Metric | Primary Focus | When to Use |
|---|---|---|---|
| Regression | MAE | Average Error | Outliers are not critical. |
| Regression | RMSE | Large Errors | Large errors are disastrous. |
| Classification | Precision | False Positives | Spam detection; cost of FP is high. |
| Classification | Recall | False Negatives | Disease diagnosis; cost of FN is high. |
| Classification | F1-Score | Balance | Imbalanced datasets. |

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# Thank You

There is no "one ring to rule them all"—context dictates the metric.

**Q&A**